Benchmark: **MA.7.AR.4.5** · Real-world proportional problems Student code: _____ Calculator: yes

Learning goal: I can solve multi-step problems with proportional relationships. (circle your level after scoring)

4	I can compare two proportional situations and justify which is the better choice.
3	I can solve a real-world problem with a proportional relationship in two or more steps.
2	I can find a unit rate.
1	I need help to start this benchmark.

1. A horse-drawn carriage travels at a constant rate. It travels $\frac{1}{2}$ mile in 10 minutes.

At this rate, how many minutes will it take the carriage to travel $2\frac{3}{4}$ miles?

2. A 3-D printer uses $\frac{3}{5}$ ounce of plastic every $\frac{1}{3}$ hour, at a constant rate.

How many ounces of plastic does the printer use in 12 hours? Write your answer as a fraction or a mixed number in simplest form.

3. A store sells ribbon at a constant price. $4\frac{1}{4}$ meters of ribbon cost $15.30.

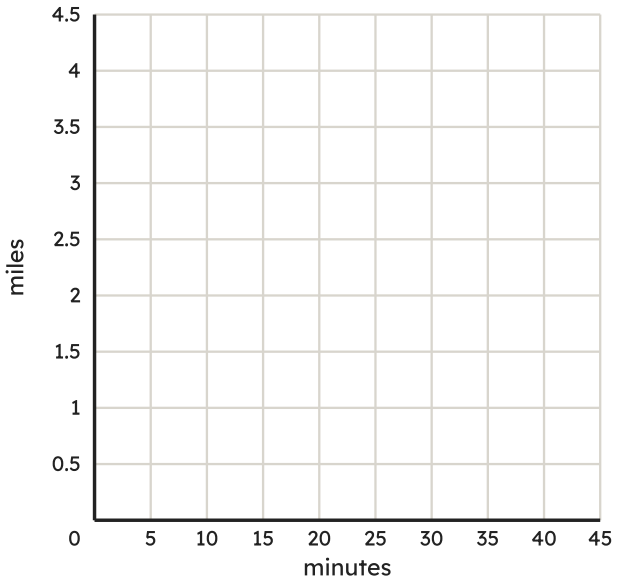
At that rate, what is the cost of $7\frac{1}{2}$ meters of ribbon?

Turn over →

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems Student code: _____

4. A carriage travels at a constant rate. After 5 minutes, it has traveled $\frac{1}{2}$ mile. The relationship between minutes and miles is proportional.

Graph the relationship on the grid. Then use the graph to find how many minutes it takes the carriage to complete a 4-mile ride.



5. Store A sells rope for \$6.30 for $2\frac{1}{4}$ meters. Store B sells the same rope for \$13.75 for 5 meters. At each store, the cost is proportional to the amount bought.

Which store is the better buy? Explain, and tell how much lower its price is per meter.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.5 · MA.7.AR.4.5 · Real-world proportional problems

Unit 6: Multi-Step Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **55 minutes**

Minutes for 1 mile: $10 \div \frac{1}{2} = 10 \times 2 = 20$ minutes.

Minutes for $2\frac{3}{4}$ miles: $2\frac{3}{4} \times 20 = 55$ minutes.

2. Regular $21\frac{3}{5} = \frac{108}{5}$ **ounces**

Unit rate: $\frac{3}{5} \div \frac{1}{3} = \frac{3}{5} \times 3 = 1\frac{4}{5}$ ounces per hour.

In 12 hours: $12 \times 1\frac{4}{5} = 21\frac{3}{5}$ ounces.

3. Regular **$27.00**

Write the lengths as decimals: $4\frac{1}{4}$ meters is 4.25 meters and $7\frac{1}{2}$ meters is 7.5 meters.

Unit price: $\$15.30 \div 4.25 = \$3.60$ per meter.

Cost: $7.5 \times \$3.60 = \$27.00$.

4. Regular **A straight line through $(0,\ 0)$ and $(5,\ 0.5)$; the 4-mile ride takes 40 minutes. Score: 1 point for the graph and the minutes; the graph alone earns 0.**

Plot $(0,\ 0)$ and $\left(5,\ \frac{1}{2}\right)$ and draw the straight line through them.

The carriage takes $5 \div \frac{1}{2} = 5 \times 2 = 10$ minutes per mile, so the line passes through $(40,\ 4)$: $4 \times 10 = 40$.

Read across from 4 on the miles axis to the line, then down to 40 minutes.

5. Rigor **Store B; its price per meter is lower, $2.75 compared with $2.80, so each meter costs less there; lower by $0.05 per meter. Score: 1 point for the store with the unit-price explanation and the amount per meter; the store alone earns 0.**

Store A: $\$6.30 \div 2.25 = \$2.80$ per meter.

Store B: $\$13.75 \div 5 = \$2.75$ per meter.

$\$2.80 - \$2.75 = \$0.05$, so Store B has the lower price, by $0.05 per meter.

Re-check strip (3 items, C-AR.4.5-R)

1. Regular **A straight line through $(0,\ 0)$ and $(12,\ 0.75)$; the 2-mile ride takes 32 minutes. Score: 1 point for the graph and the minutes; the graph alone earns 0.**

Plot $(0,\ 0)$ and $\left(12,\ \frac{3}{4}\right)$ and draw the straight line through them.

The carriage takes $12 \div \frac{3}{4} = 12 \times \frac{4}{3} = 16$ minutes per mile, so the line passes through $(32,\ 2)$: $2 \times 16 = 32$.

Read across from 2 on the miles axis to the line, then down to 32 minutes.

2. Regular **24 minutes**

Minutes for 1 mile: $8 \div \frac{2}{3} = 8 \times \frac{3}{2} = 12$ minutes.

Minutes for 2 miles: $2 \times 12 = 24$ minutes.

3. Regular **9 miles**

Unit rate: $\frac{1}{2} \div \frac{1}{6} = \frac{1}{2} \times 6 = 3$ miles per hour.

In 3 hours: $3 \times 3 = 9$ miles.

Comparison strip (2 items, C-AR.4.5-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **D. 10.5 gallons**

Unit rate: $85 \div 2.5 = 34$ miles per gallon.

$357 \div 34 = 10.5$ gallons.

$2.5 \times 85 \div 357$ turns the ratio upside down, $357 \div 2.5$ treats gallons as miles per gallon, and $2.5 + 357 \div 85$ adds the scale factor instead of multiplying by it.

2. Regular **$16.20**

Write the lengths as decimals: $3\frac{3}{4}$ meters is 3.75 meters and $6\frac{3}{4}$ meters is 6.75 meters.

Unit price: $\$9.00 \div 3.75 = \$2.40$ per meter.

Cost: $6.75 \times \$2.40 = \$16.20$.

Reteach plan — C-AR.4.5

MA.7.AR.4.5 · Real-world proportional problems

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

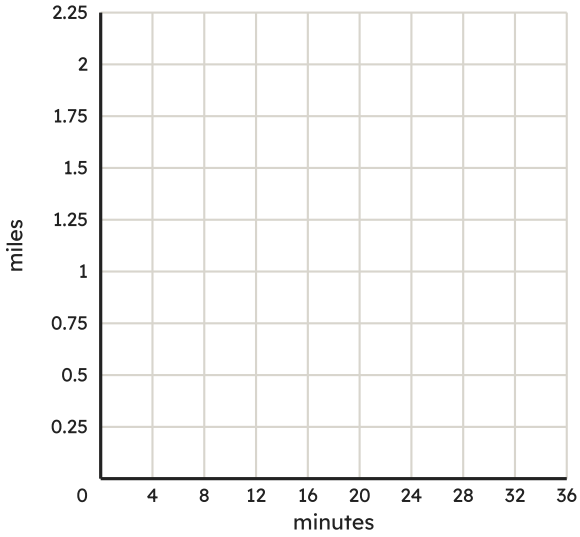
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.5-R

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems Student code: _____ Calculator: yes

1. A carriage travels at a constant rate. After 12 minutes, it has traveled $\frac{3}{4}$ mile. The relationship between minutes and miles is proportional.

Graph the relationship on the grid. Then use the graph to find how many minutes it takes the carriage to complete a 2-mile ride.



2. A horse-drawn carriage travels at a constant rate. It travels $\frac{2}{3}$ mile in 8 minutes.

At this rate, how many minutes will it take the carriage to travel 2 miles?

3. A hiker walks $\frac{1}{2}$ mile every $\frac{1}{6}$ hour, at a constant rate.

How many miles does the hiker walk in 3 hours?

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2)

C-AR.4.5-X

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems

Student code: _____ Calculator: yes

1. A car uses gas at a constant rate. It travels 85 miles on 2.5 gallons of gas.

How many gallons of gas are needed to travel 357 miles?

A. 142.8 gallons

B. 6.7 gallons

C. 0.6 gallons

D. 10.5 gallons

2. A store sells ribbon at a constant price. $3\frac{3}{4}$ meters of ribbon cost $9.00.

At that rate, what is the cost of $6\frac{3}{4}$ meters of ribbon?

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.4.5-X

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems

Student code: _____ Calculator: yes

1. A car uses gas at a constant rate. It travels 85 miles on 2.5 gallons of gas.

How many gallons of gas are needed to travel 357 miles?

A. 142.8 gallons

B. 6.7 gallons

C. 0.6 gallons

D. 10.5 gallons

2. A store sells ribbon at a constant price. $3\frac{3}{4}$ meters of ribbon cost $9.00.

At that rate, what is the cost of $6\frac{3}{4}$ meters of ribbon?

Score: ____ **/ 2** → **copy to your progress strip**